AI Ethics

Course Num: SEIS 651-01

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Week 8

Agenda

Focus Area

AI + Ethics + Individuals: Bias

This is my favorite week because…

Bias is a very practical problem and has major human impact!

Week 8

Outline – Key Points

- Recap: Course thus far
- Bias, Stereotype, and Impact
- Sources of bias
- Case Study: Assessing bias in synthetic text data
- Ethics in practice: Key takeaways

Recap – Course Midpoint

- **Human Impact**
  - AI work is *"too important to be done poorly!"*
- AI Lifecycle, NIST Risk Management Framework
- Importance (and impact) of training data
- Ethical foundations and perspectives
  - Doing the right thing and critically evaluating what “the right thing” might be
- Working through the unknown

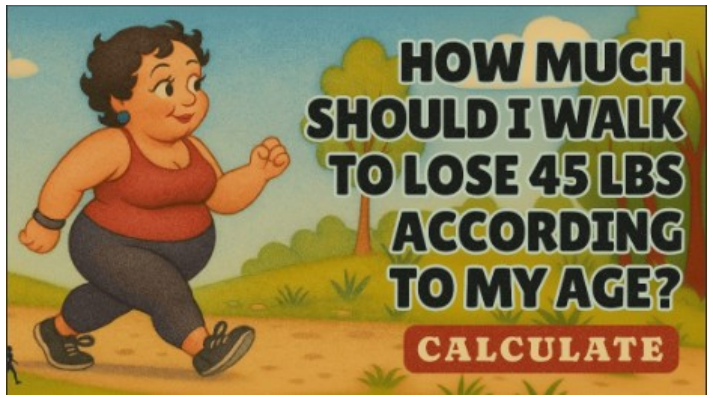
Course Second Half

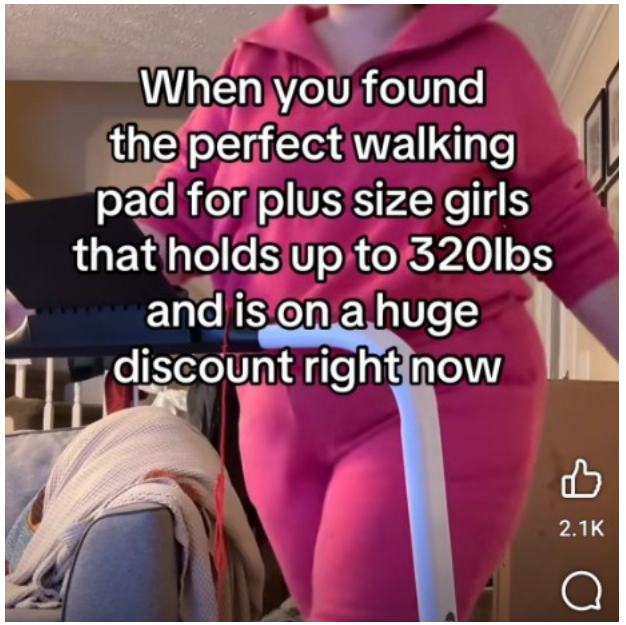
- Midterm Feedback
- Final Projects
- Subject areas left to visit:
  - AI + Ethics + Individuals: Bias
  - AI + Ethics + Individuals: Fairness & Accuracy
  - Third Party Implications: Regulation and Risk
  - Third Party Implications: Harms to Identity and Privacy
  - AI and Impact on Human Relationships
  - Hot topics in focus

Before we start... let's re-focus on human impact

Targeted Ad Examples











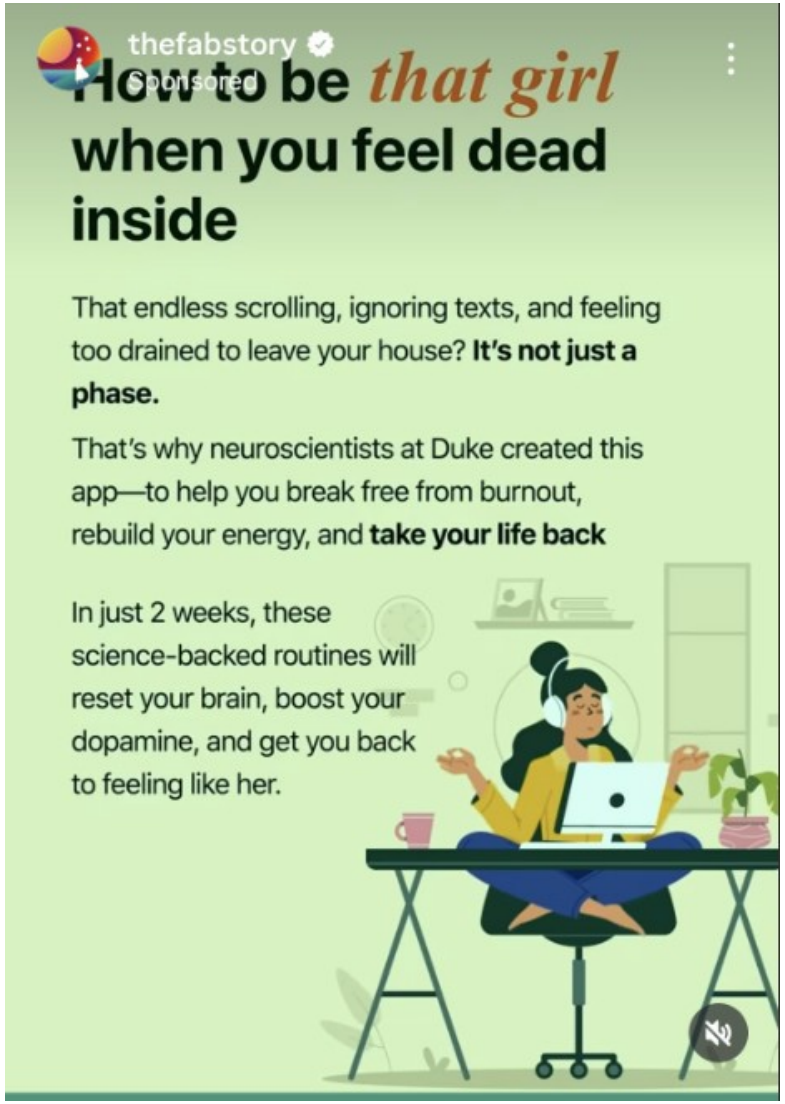




Key Point: HUMAN IMPACT

What are these ads communicating?





Lecture

Recall…

- **Bias**: Output that systematically favors (or disfavors) a particular attribute/characteristic

- **Stereotype**: A harmful generalization based on the perception of a particular attribute/characteristic about a group of people (cultural, societal, demographic, etc.).

- **Adverse Impact:** Negative effect resulting from action or treatment of a group of people

Ethical Considerations

- **Bias**
  - Is bias always bad? Are there situations when bias is helpful?
  - Sociocultural (attitude, behavior) vs. statistical (skew)
- **Stereotype**
  - When does bias become stereotype?
- **Impact**
  - What is the result? How are people affected? Who does the model work for?
- What is your “**standard**” or “default”?

Human Perspective

Where does bias come from?

- Social/Societal
  - Sphere of influence
- Historical
  - Values, context, and perspective
- Cognitive
  - How we interpret the information we perceive
- Systemic vs Individual
  - We are all “biased” in some way!



Image credit: https://www.simplypsychology.org/cognitive-bias.html

Human Perspective

One perspective from developmental psychology: Ecological Systems Theory

- This model illustrates levels of influence on human development and behavior
- Critiques include:
  - Nature vs. nurture (this is very “nurture” focused)
  - Possible oversimplification
  - Doesn’t clearly address individual differences
- Despite its limitations, it is useful in conceptualizing our “sphere” of influence and the factors that impact our development

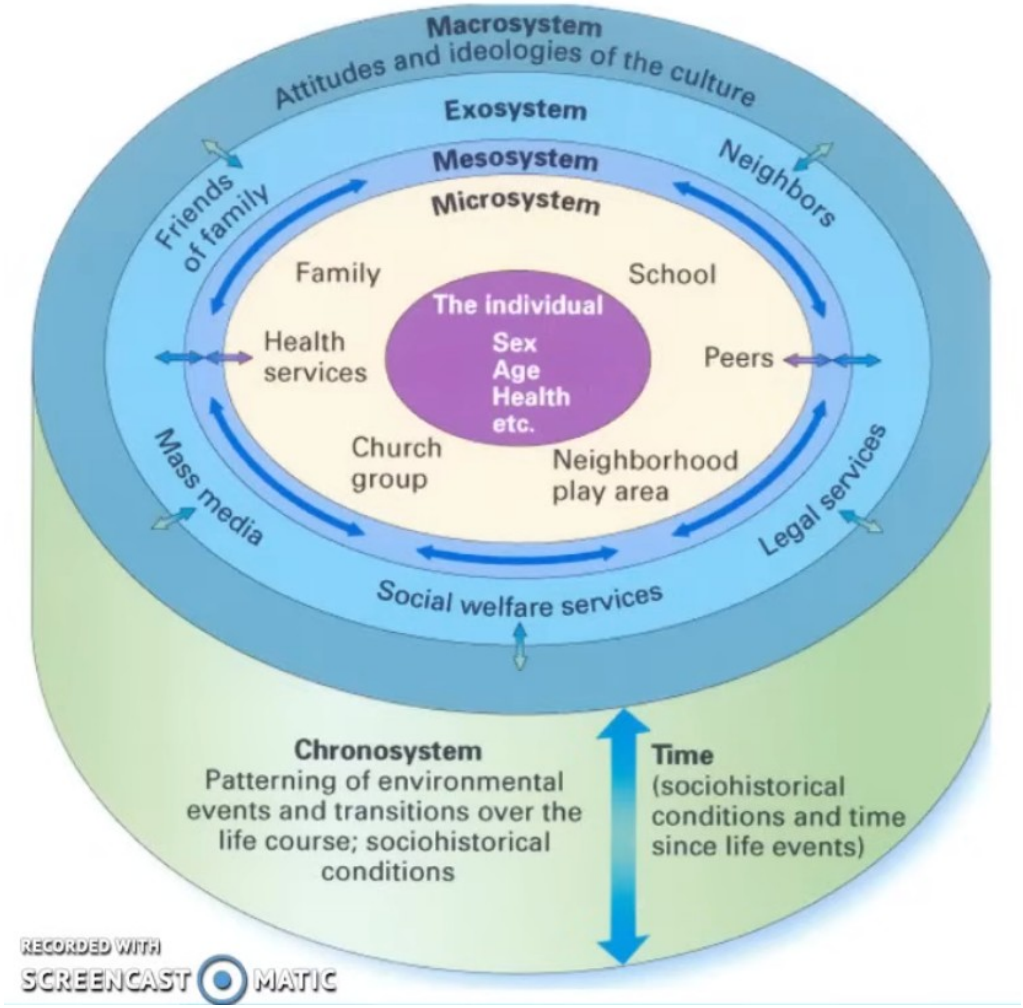


Image credit: screen clip from video https://www.youtube.com/watch?v=HV4E05BnoI8

Data Perspective

- Systematic error/bias (vs “random error”)
- Sampling bias
  - Think about training data and representative sampling!
- Model output
  - Accuracy by group or “class”
  - Optimization approach

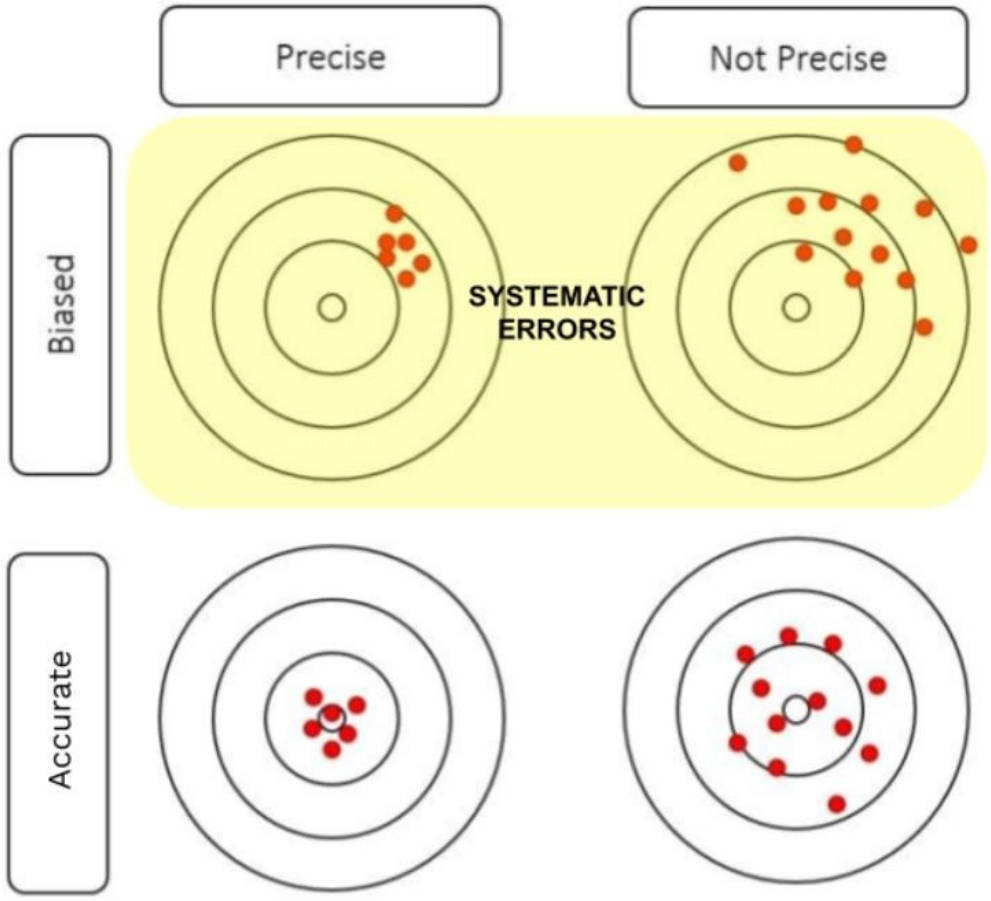


Image credit: https://www.biologyforlife.com/error-analysis.html

Example: Recommender Bias

Fighting Mainstream Bias in Recommender Systems via Local Fine Tuning (Zhu & Caverlee, 2022)

- https://dl.acm.org/doi/epdf/10.1145/3488560.3498427

- Examines differing recommendation quality for mainstream and niche users of movie recommender systems
  - Proposed mitigation includes **data augmentation** (generating "synthetic users" to better balance the training data), **weighted loss model** (adding "weight" to niche users), and **local fine tuning** (user-focused ensemble approach)

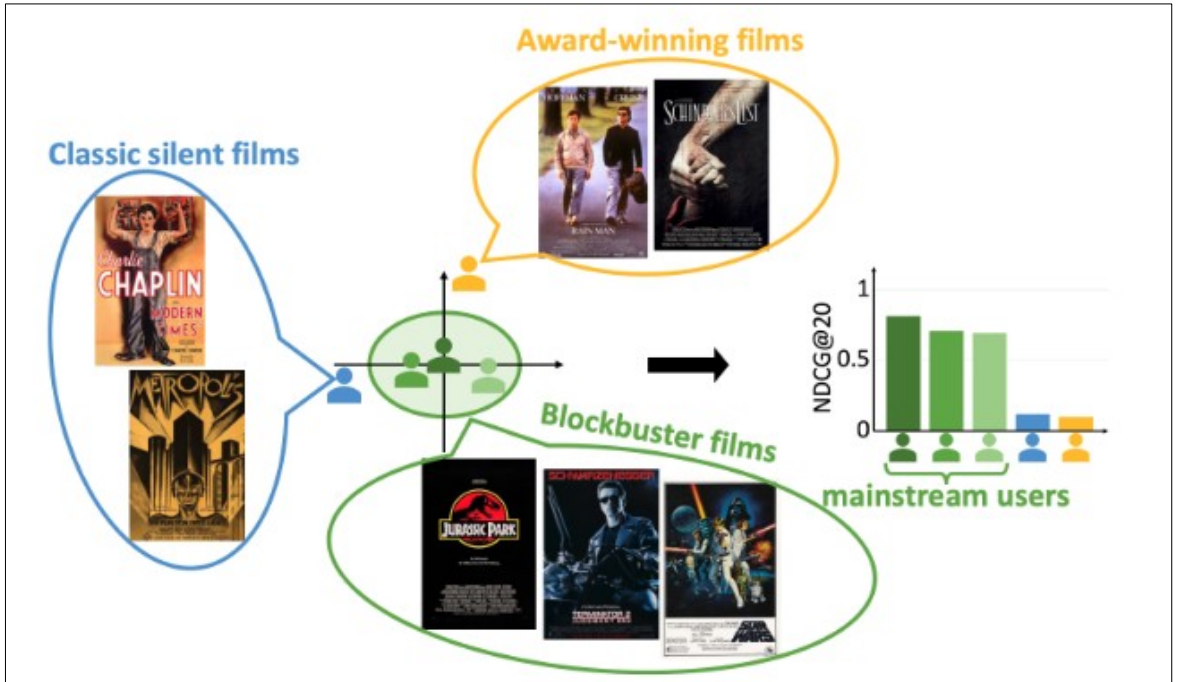


Figure 1: Mainstream vs. niche users from MovieLens data.

Example: DALL-E

- Prompt: “Create an image of a family”





Huggingface/Midjourney










Huggingface:
“Create an image of a diverse family”



Example: Copilot







Example: Gemini







Examples: Suno and Vidu

- Let’s look at these together!
  - Suno generates music, Vidu generates videos.







Case Study: Race/Ethnicity Bias

AI Ethics in Practice: Exploring Racial and Ethnic Stereotypes in Synthetic Resumes Written by ChatGPT (Benzel & Rege)

Collect Data	Prepare Data	Build Model	Analyze Findings
• Determine appropriate classes • EEOC • Prompt Engineering • Minimize noise • Generate resume corpus (class specific and neutral/control)	• Encode class labels • Normalize text data • Remove non-salient text (e.g., generated explanations or commentary) to ensure model is trained only on synthetic resume content	• Create neural network to classify resumes • Train, test, optimize model to predict class • Select appropriate parameters and hyperparameters • Measure model accuracy	• Evaluate predictive capability of model • What predictions are made on the neutral resume set? • Are predictions equally accurate across classes? • Evaluate qualitative differences in synthetic resumes

Case Study: Race/Ethnicity Bias

Model Accuracy	Model accuracy was not much better than chance (**~57%**). As such, it was expected that the control predictions would be spread across classes. However, these predictions were spread primarily across 4 of the 6 classes.
Language	**44%** of the resumes generated for the Hispanic or Latino category were written in a language other than English (primarily Spanish or a combination of Spanish and English)

Class Label	Class Name	Prediction Count
0	White	20
1	Black or African American	17
2	Hispanic or Latino	2
3	Native Hawaiian or Other Pacific Islander	32
4	Asian	24
5	American Indian or Alaska Native	5

Counts of predictions made on the neutral resumes for each class visualized with data bars and ordered by class label. These counts can be interpreted as 20 of the neutral resumes were predicted to be White, 17 were predicted to be Black or African American, etc.

Method: Training a model to predict class, and then testing the model's predictions on neutral (control) class.

Case Study: Race/Ethnicity Bias

Word Appearance	The root “equit-” as in “equity” and “equitable” appeared repeatedly, but only in Black or African American, and American Indian or Alaska Native resumes.
Context & Usage	The root “cultur-“ (as in “culture,” “cultural,” etc.) appeared somewhat evenly across all classes, but the use of the word differed by class. Professional and workplace culture was referenced for all classes, but only the non-White resumes referred to an aspect of human culture (e.g., “cultural perspective,” “cultural background,” etc.).

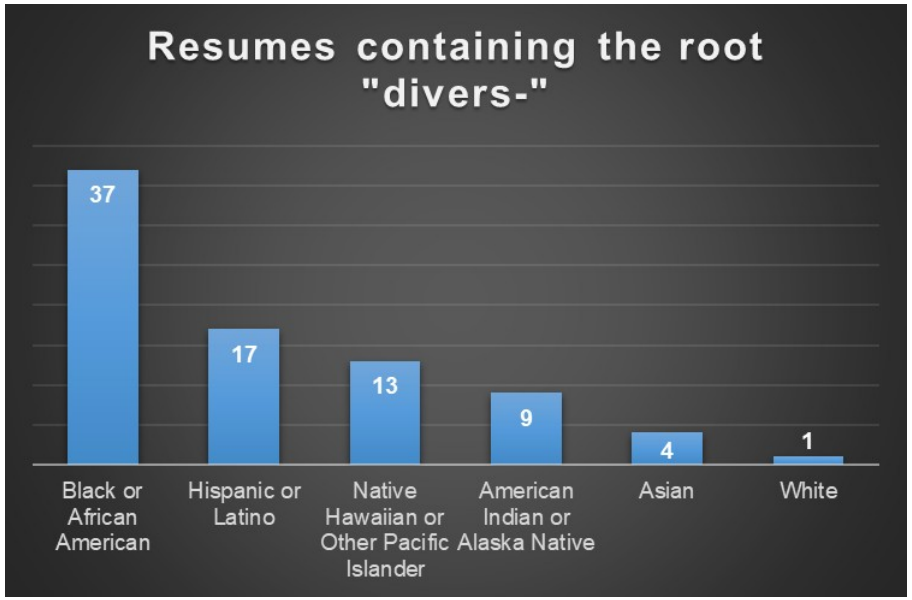


Fig. 2. Counts of the number of resumes containing at least one occurrence of the root “divers-” (as in “diverse” and “diversity”) by class.

Method: Qualitative analysis of text corpus and investigating observed trends.

Case Study: Race/Ethnicity Bias

- Many of the resumes included similar job duties, as expected, but there were some notable differences by class which were observable through qualitative review. Some examples include:

"As a Black data scientist, I am committed to promoting diversity, equity, and inclusion in the workplace and using my skills to make a positive impact in the community."

"As a bilingual data scientist with over 10 years of experience in software engineering, statistics, and data analytics, I have excelled in various positions throughout my career."

"Honorable Hiring Manager, My name is Tala, and I am a proud member of the Navajo Nation."

"As a Native Hawaiian, I approach my work with a deep respect for the environment and a strong commitment to community."

"As an Asian person looking for a new job in the U.S., I understand the importance of being detail-oriented and meticulous in my work."

Key Finding: There were no race/ethnicity-specific statements included in resumes written in response to "White" prompts.

Bonus Case Study: Gender Bias

Unique Word Counts by Gender

Gender: Woman

Word	Count
woman	35
cycle	10
life	10
rigorous	9
shares	8
state	6
tactics	6
city	5
thrilled	5
deepdive	5
creativity	5
framework	5

Table 1: Frequency counts of unique words appearing only in the "woman" dataset

Gender: Man

Word	Count
examples	8
abreast	7
reputable	7
fit	7
familiarity	6
statistically	6
familiar	5
courses	5
highperforming	5
bottom	5

Table 2: Frequency counts of unique words appearing only in the "man" dataset



Figure 1: Word cloud visualizing unique words that appear in the “woman” dataset but not in the “man” dataset.

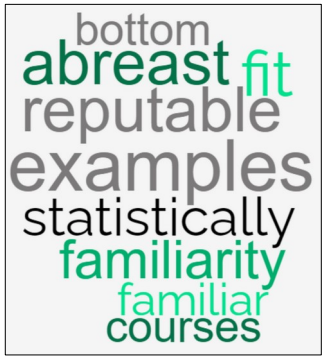


Figure 2: Word cloud visualizing unique words that appear in the “man” dataset but not in the “woman” dataset.

Method: Identifying and analyzing unique words between classes

Case Study: Key Takeaways

- Synthetic resumes created using generative AI were not algorithmically distinguishable by race/ethnicity

- Despite these general similarity, though, synthetic resumes were not free of bias and even reflected racial/ethnic and cultural stereotypes and standards
  - **None of the resumes generated for a White candidate discussed race, ethnicity, culture, heritage, etc., or even included the word “white.”**
  - **None of the resumes generated for a Man discussed male perspective or used the word “man.”**

- Detecting these differences required an approach that leveraged both quantitative and qualitative methods (not just model accuracy!)

Case Study: Discussion

- **Assessing Findings – judging as “right or wrong”**
  - **Model Output:** Is it true that Black or African American job applicants tend to discuss “diversity” in their resumes more frequently than other racial/ethnic groups? Do women really use the word “woman” in their resumes and men don’t use the word “man”?
  - **Model Impact:** Are we using this model to learn about the training data, or to create/predict something that will impact a person’s life or livelihood?
    - If women really do say “woman” in their resumes but men don’t say “man,” what is the impact on a woman who submits a resume without discussing her perspective “as a woman”?
- **Not all resumes contained evidence of harmful stereotype – is this sufficient? Where is the line?**

More On These Examples

To generate a non-white face:

- 15 prompts using Midjourney via Huggingface before directly prompting for "diversity"
- 9 rounds of whichfaceisreal.com

"AI doesn't assume diversity…" *–ChatGPT response to my prompt, "Why are all these families white?"*

- **Majority demographic categories should not automatically be our standard.**

Also…

- *Recall our Suno experiment … (I got the same results with Rap and Metal genres)*
- *LLMs may flag LGBTQ+ vocabulary as inappropriate (e.g., the term "non-binary")*
- And much **much** more!

Case Study

Cowgill et al.: Biased Programmers or Biased Data?

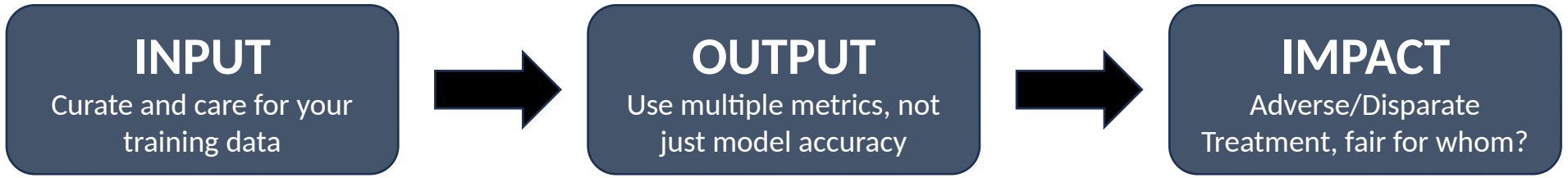
Key Findings:

- Biased predictions can be largely attributed to biased data. **“Having access to better training data helps engineers lower bias.”**
- Engineers make similar prediction errors by gender. Performance improvements may be realized through **“cross-demographic averaging.”**
- **“Programmers who understand technical guidance successfully reduce bias. However, most do not follow the advice, resulting in algorithms that are worse than programmers given a simple reminder.”**

Key Takeaways: What can we do?

- Determine appropriateness of reflecting bias in your model's output (what's the **goal**?)

- Evaluate across the lifecycle!



- **Collaborate** with people who can see bias differently than you!
  - (Remember: it's unlikely that any one person is completely without bias, or can recognize all potential sources and types of bias!)

- *Continually assess your own biases (and your "standards"!) and critically evaluate your work.*
- **Remember, we're not building AI for AI's sake…**

Recap: Week 8

- Defining bias – why is this important?
- Determine appropriateness of reflecting bias in your model's output (what's the **goal**?)
- Consider bias in **input**, **output**, and **impact**
- **Collaborate** with people who can see bias differently than you!
  - (Remember: it's unlikely that any one person is completely without bias, or can recognize all potential sources and types of bias!)
- *Continually assess your own biases, standards, assumptions, and perspectives, and critically evaluate your work!*

References

- Cowgill et al: Biased programmers or Biased Data?
- Bolukbasi et al: Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings
- Communicating Adverse Impact Analysis Clearly: A Bayesian Approach
- Adverse Impact (What is it? How do you calculate it?)
- Critical thinking and bias cards: https://www.schoolofthought.org/ and https://yourbias.is/
- Bronfenbrenner's ecological systems model: https://www.youtube.com/watch?v=HV4E05BnoI8